Magic Millipedes Exhibit

System Requirements Document v1.0.0

Concordia University Irvine

Project Management (ENGR 210)
Software Engineering (CSC 322)

Introduction

Purpose

This document defines the formal system requirements for the *Magic Millipedes Exhibit*, an interactive museum installation featuring live bioluminescent and fluorescent millipedes. It will be deployed in a public museum, so the requirements emphasize safety, robustness, and recoverability in addition to functional correctness.

Scope

The system includes both hardware and software components that together control visitor flow, lighting, audio narration, and interaction inside the Millipedarium. The system governs four identical interactive rooms, each operating independently but following the same specifications.

Definitions

- **Room**: One of enclosed visitor spaces inside the Millipedarium.
- **Presentation Cycle**: A two-minute audio/light sequence initiated by a button press.
- **Default State**: The idle state of a room when no presentation is active.

System Overview

The Millipedarium is a 16 × 12 × 8 ft structure serving as the focal point of the exhibit. Visitors enter individual rooms through exterior doors. The rooms operate independently of each other using identical hardware and software configurations. Each room contains:

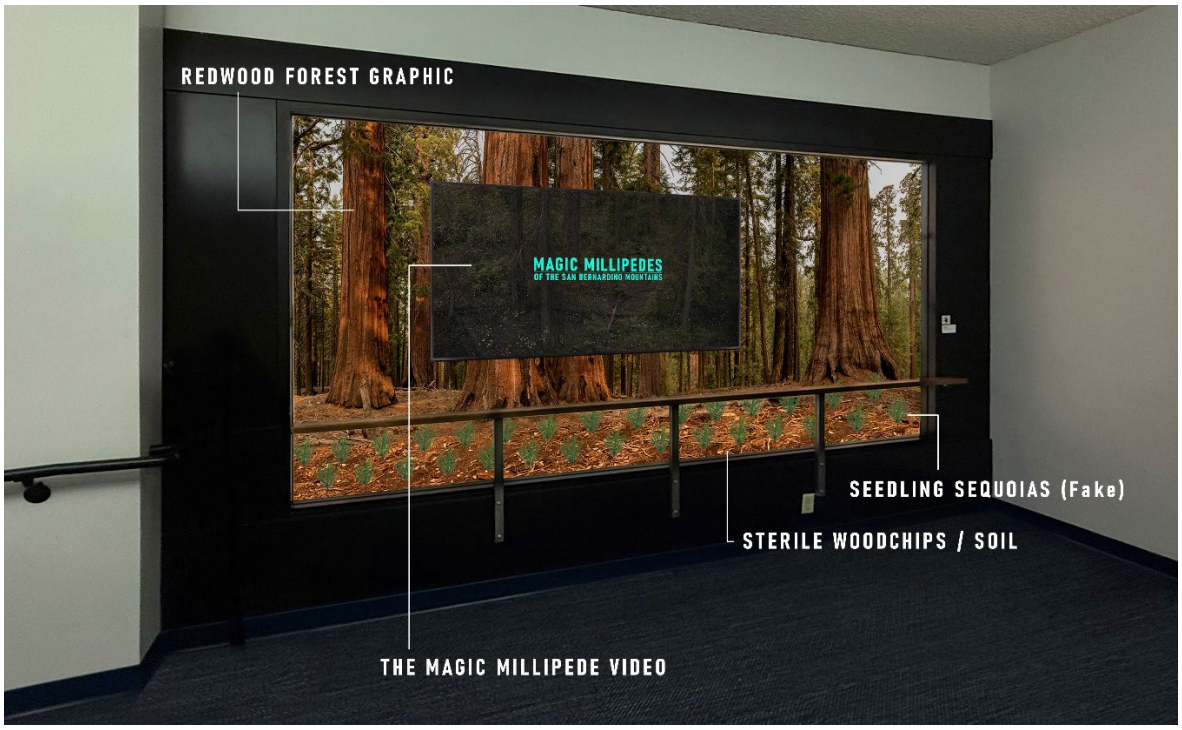
1. A glass-enclosed habitat containing live millipedes.
2. A UV light source within the glass-enclosed millipede habitat.
3. Interior and exterior indicator lights that communicate room availability.
4. A push-button trigger to begin the presentation.
5. An audio speaker to narrate scientific information.

Millipede care systems (temperature, humidity, feeding) are outside the scope of this document.

The system coordinates these components to create a timed educational experience while managing visitor access.

MAGIC MILLIPEDES EXHIBIT

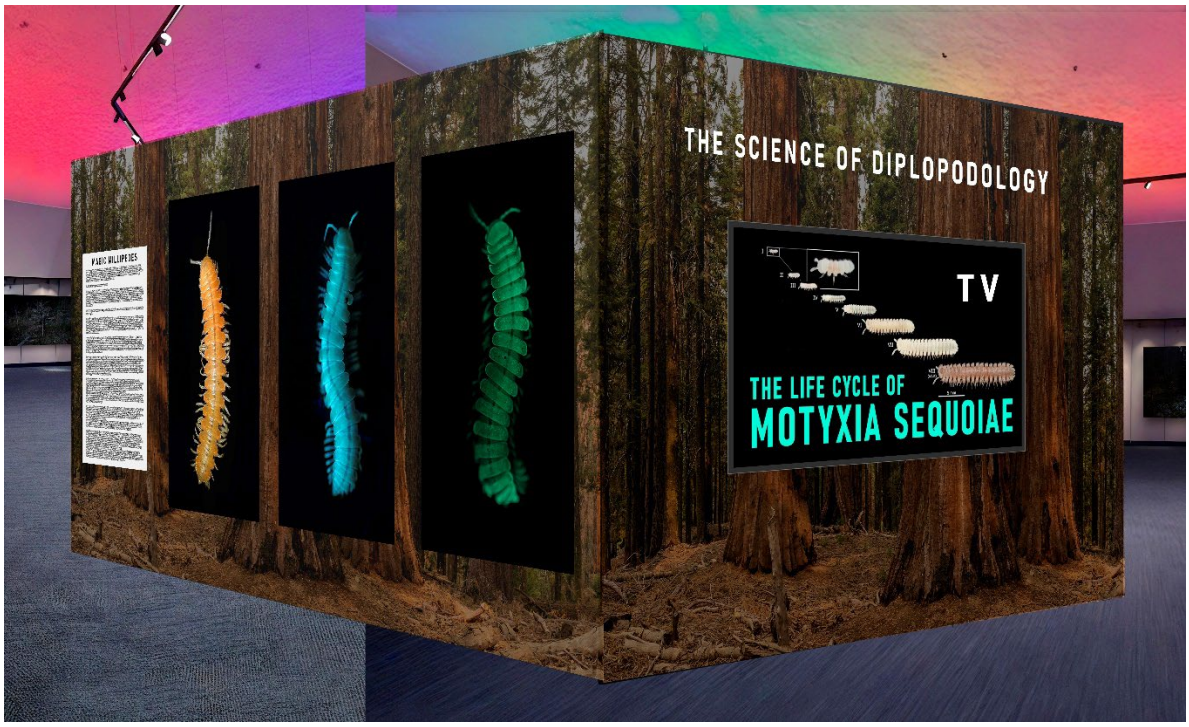
ENTRANCE



Main Room

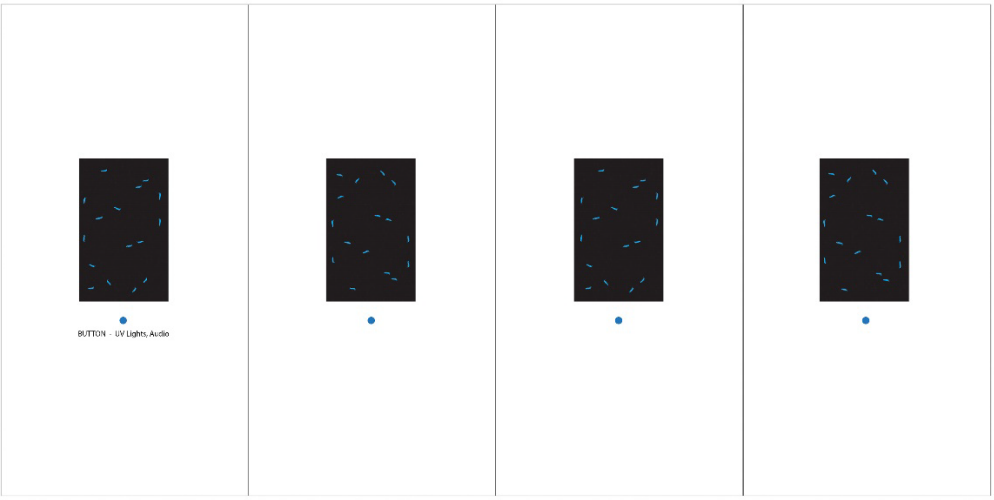
Millipedarium Left Side/Front

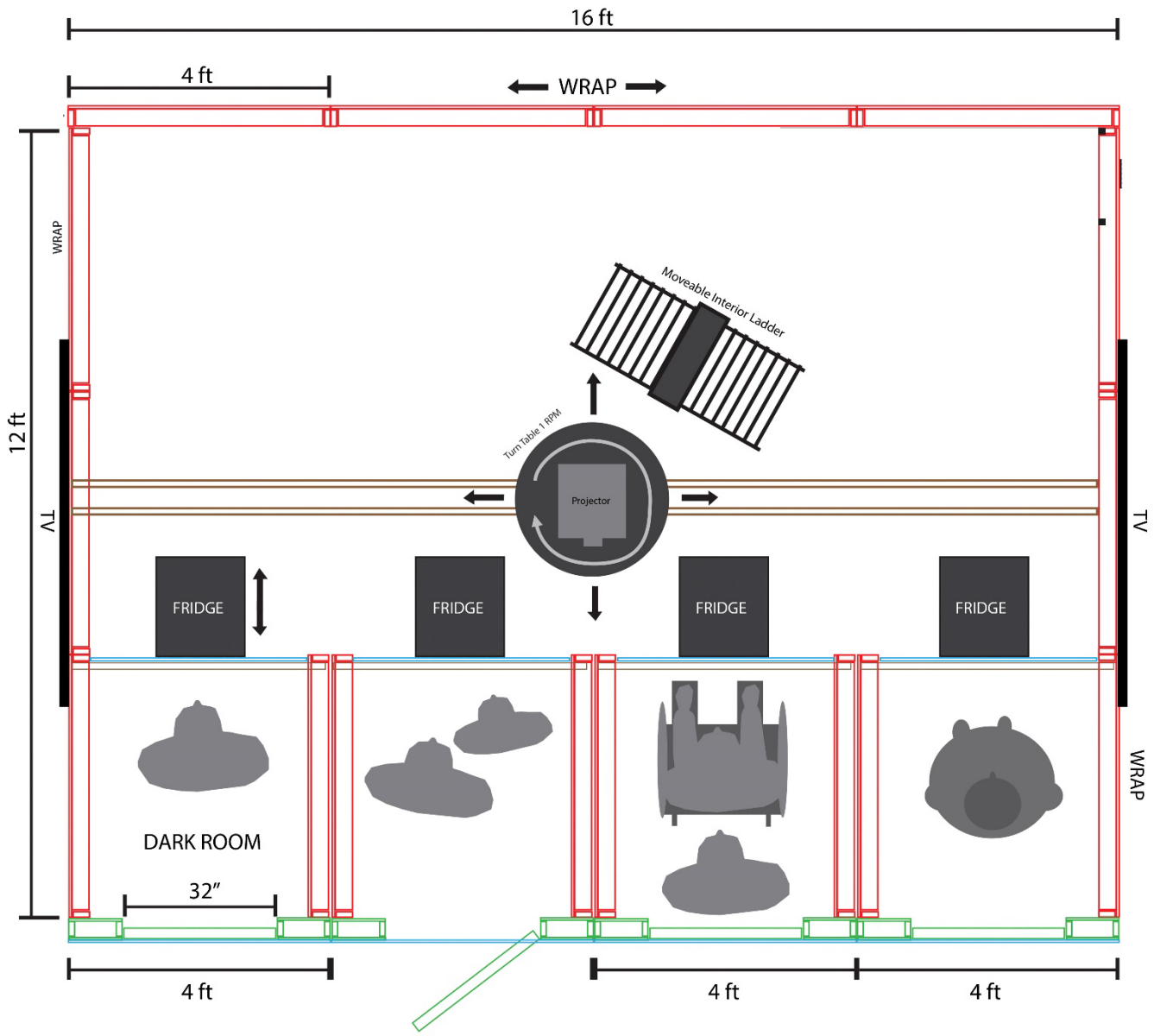
Millipedarium Right Side/Front



Millipedarium Back

INTERIOR ROOMS



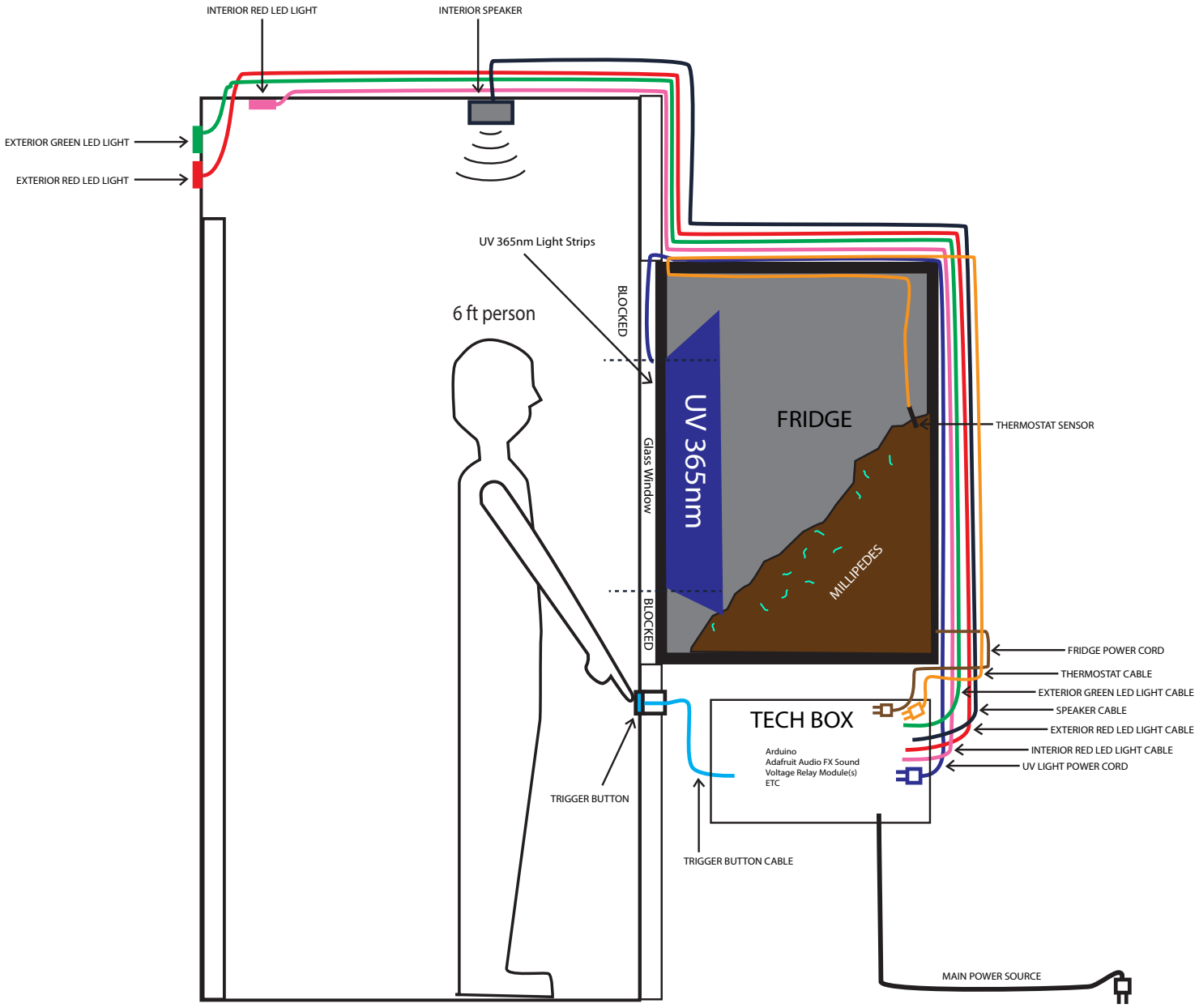


<u>The Millipedarium</u>

The Millipedarium (16 x 12 x 8 ft) is a structure serving as the focal point of the exhibit. The exterior walls will feature large photographs, 2 video screens, educational information, and infographics. The outside of the structure will be wrapped in a redwood forest graphic. The rear wall will have 4 doors leading to separate rooms inside.

Above each door will be a red and green light, which lets visitors know when to enter or wait. Each room will contain a display of live, bioluminescent/fluorescent millipedes behind glass. Inside the room, visitors will see a button below the glass. When the button is pushed, a 2-minute audio/light presentation begins.

Below is a diagram of the button/light/audio presentation inside each room.


INTERIOR RED LED LIGHT
INTERIOR SPEAKER
EXTERIOR GREEN LED LIGHT
EXTERIOR RED LED LIGHT
UV 365nm Light Strips
6 ft person
BLOCKED
UV 365nm
FRIDGE
THERMOSTAT SENSOR
Glass Window
MILLIPEDES
BLOCKED
FRIDGE POWER CORD
THERMOSTAT CABLE
EXTERIOR GREEN LED LIGHT CABLE
SPEAKER CABLE
EXTERIOR RED LED LIGHT CABLE
INTERIOR RED LED LIGHT CABLE
UV LIGHT POWER CORD
TECH BOX
Arduino
Adafruit Audio FX Sound
Voltage Relay Module(s)
ETC
TRIGGER BUTTON
TRIGGER BUTTON CABLE
MAIN POWER SOURCE

Feature 1: Room Availability Subsystem

This feature governs how visitors determine when it is appropriate to enter or wait for a room. The system communicates room availability exclusively through exterior visual indicators, avoiding the need for staff intervention or written instructions.

Functional Requirements

- **F1.1** The system shall display a green exterior indicator light when a room is available for entry.
- **F1.2** The system shall display a red exterior indicator light when a presentation cycle is in progress (see Feature 2: Presentation Control System).

Non-Functional Requirements

- **NF1.1** Indicator lights shall be visible from a minimum distance of 10 feet under typical gallery lighting conditions.
- **NF1.2** Indicator lights shall operate independently for each room.

Acceptance Criteria

- **AC1.1** Visitors must correctly identify whether to enter or wait for a room based solely on the exterior indicator lights in 9 out of 10 observed interactions.

Feature 2: Presentation Subsystem

This feature defines the behavior of the interactive experience initiated by a visitor. The presentation control system coordinates light and audio outputs according to a fixed two-minute sequence. The system should be designed to educate visitors while protecting the millipedes.

Functional Requirements

- **F2.1** The system shall initiate a presentation cycle when the trigger button is pressed.
- **F2.2** The presentation cycle shall last exactly two minutes.
- **F2.3** The system shall execute the following timed phases during each presentation cycle:
  - **0:00–0:30**: Audio narration begins; interior red light remains on; UV light remains off. This phase ends when the first, instructional audio narration finishes.
  - **0:30–1:30**: Interior red light turns off; UV light turns on; audio narration continues with a biological description of the millipedes.
  - **1:30–2:00**: UV light turns off; room remains dark; audio plays soft classical music.
- **F2.4** While in the presentation cycle, button presses shall have no effect on the system.
- **F2.5** At the conclusion of the cycle, the system shall return the room to the Default State.

Non-Functional Requirements

- **NF2.1** Timing accuracy for all phases shall be within ±1 second.
- **NF2.2** Audio playback shall be clearly audible to occupants inside the room without being disruptive outside the room. (Design Note: Typical indoor speech intelligibility occurs between 60–70 dB SPL at the listener's position; use this as a reference only).

Acceptance Criteria

- **AC2.1** A presentation must execute all phases in the correct order and duration for 10 consecutive cycles without error.

Feature 3: Lighting Subsystem

The lighting system provides both functional and educational illumination, including visitor guidance lights and UV illumination used to reveal millipede fluorescence.

Functional Requirements

- **F3.1** The system shall control exterior red and green LED indicators for each room (see Feature 1: Room Availability Subsystem).
- **F3.2** The system shall control an interior red LED light used during room entry and early presentation phases (see Feature 2: Presentation Control Subsystem).
- **F3.3** The system shall control an interior UV light source capable of illuminating millipede fluorescence. The UV light shall be externally sourced and treated as a fixed component with defined electrical characteristics (see appendix).

Non-Functional Requirements

- **NF3.1** The UV light shall be from realUV 365 nm LED light strips. The specified UV light is a contractual requirement due to biological, safety, and wavelength constraints.
  https://store.waveformlighting.com/products/real-uv-led-strip-lights?variant=12339996721254
  https://store.waveformlighting.com/products/realuv-dc-power-supply-for-led-strip
- **NF3.2** The UV light shall be safe for enclosed operation within a humid, refrigerated environment.
- **NF3.3** Lighting transitions shall appear instantaneous to visitors.

Acceptance Criteria

- **AC3.1** Millipede fluorescence must be clearly visible during the UV illumination phase to an average visitor.

Feature 4: Audio System

The audio system delivers synchronized educational narration and soft background music that supports the visual presentation without interfering with neighboring rooms or gallery spaces.

Narration audio will be provided to the engineering design team in two separate files. The first file will include the instructions and countdown timer used in the first 0:30 seconds. The second audio file will include the educational content for the 0:30-1:30 portion of the presentation cycle.

Functional Requirements

- **F4.1** The system shall play a recorded educational narration synchronized with the presentation timeline (see Feature 2: Presentation Control Subsystem).
- **F4.2** The system shall support countdown narration prior to UV activation.
- **F4.3** The system shall play soft classical music during the final phase of the presentation.

Non-Functional Requirements

- **NF4.1** Audio content shall be replaceable without modifying system logic.
- **NF4.2** Audio playback shall automatically stop at the end of each presentation cycle.

Acceptance Criteria

- **AC4.1** Visitors must hear all narration segments clearly and in the correct sequence during a full presentation cycle.

Feature 5: Control Electronics

The control electronics coordinate all inputs and outputs within a room, serving as the logical backbone of the system. The electronics support basic operational monitoring to assist with maintenance and exhibit usage assessment. Solutions that rely on exposed or stretched connections through visitor space are prohibited.

Functional Requirements

- **F5.1** The system shall use a central control unit within each room to coordinate inputs and outputs.
- **F5.2** The system shall accept a momentary button press as a trigger input.
- **F5.3** The system shall control AC-powered and low-voltage components through appropriate electrical interfaces.
- **F5.4** The system shall count the number of presentation cycles it begins.

Non-Functional Requirements

- **NF5.1** The trigger button shall be obvious and accessible to children and adults alike without bending down or reaching up from a normal, restful standing position.
- **NF5.2** Control electronics shall operate continuously for a minimum of 10 hours per day.
- **NF5.3** AC-powered components shall be electrically isolated from low-voltage control electronics and all user-accessible inputs.
- **NF5.4** Electrical components shall comply with all applicable safety standards.
- **NF5.5** The system shall have a manual reset that is only accessible to maintenance personnel and can be initiated without any specialized tools.
- **NF5.6** The momentary pushbutton shall be rated to withstand repeated high-force activation without mechanical failure, consistent with unsupervised public use by children.
- **NF5.7** The presentation counter shall persist on power loss.
- **NF5.8** The counter shall be retrievable by engineers with physical access to the system.

Acceptance Criteria

- **AC5.1** The control system must run continuously for one full exhibit day without requiring manual reset.

Feature 6: Operational Robustness

This feature defines how the system behaves under abnormal or unexpected conditions such as power loss, component failure, or improper user interaction. The exhibit will be deployed in a public museum with live millipedes and unsupervised visitors, so the system must fail safely, recover predictably, and avoid hazardous or disruptive behavior.

Functional Requirements

- **F6.1** Upon initial power application, the system shall initialize each room to the Default State.
- **F6.2** The system shall return each room to a safe Default State following power loss or system restart.
- **F6.3** The system shall prevent continuous UV illumination beyond the defined presentation duration.
- **F6.4** The system shall safely ignore or handle repeated, rapid, or malformed trigger inputs without entering an unsafe or undefined state.

Non-Functional Requirements

- **NF6.1** The system shall fail in a state that does not expose visitors or millipedes to unsafe lighting, audio, or electrical conditions.
- **NF6.2** The system shall operate safely without requiring continuous human supervision during public exhibit hours.

Acceptance Criteria

- **AC6.1** Power shall be interrupted and restored during an active presentation, and the system shall return to the Default State within 60 seconds.
- **AC6.2** Repeated button presses during an active presentation shall not extend or restart the presentation.

Assumptions and Dependencies

- The system assumes that room doors are closed during normal presentation operation.
- Routine maintenance, inspection, and repair will occur outside of public exhibit hours.
- Maintenance personnel will have physical access to control electronics and service channels without entering occupied visitor space.

Environmental and Safety Constraints

- The system shall not expose visitors or millipedes to unsafe electrical components under any operating condition.
- The system shall not expose millipedes to unsafe UV illumination durations or wavelengths.
- The system shall maintain compatibility with refrigerated enclosures used to house live millipedes.
- All interactive elements shall be accessible to visitors of varying heights and abilities.

This document was originally written by Ben Price
The format was modified using Generative AI
Reviewed and edited by Prof. Tallman, Dr. Lane, and Ben Price
Reviewed by Matthias Bruggeman of Concordia Writing Studio
Final edits by Prof. Tallman

End of Requirements Document

realUV™ LED Strip Lights

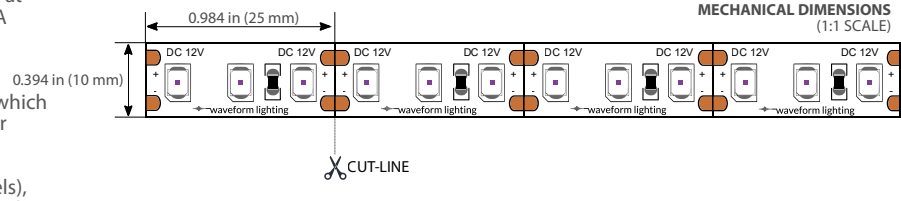
PN: 7021

Waveform Lighting's realUV™ LED strip lights provide high power ultraviolet light at 365 nanometers or 395 nanometers. These wavelengths are considered true UV-A wavelengths, and are the optimal wavelengths for activating and observing fluorescence and other UV-A phenomenon.

The back side of the LED strip includes pre-applied 3M VHB® double-sided tape, which provides a simple but extremely strong adhesive mounting method for all of your projects.

The LED strips are 16.4 feet (5.0 meters) in length (also available as short 3.2 ft reels), and are conveniently reeled for quick and easy application, and can be cut to length every 1-inch (25 mm) with just a pair of scissors.

MECHANICAL DIMENSIONS
(1:1 SCALE)



PRODUCT AND FEATURES

- Available in either 365 nm or 395 nm
- The light emitted from the 365 nm version is largely invisible, but a miniscule amount of visible light is also emitted and appears as a very dim bluish-white light.
- The 395 nm version emits a portion of its output energy in the visible wavelength range and appears as a dim, violet light.
- Available in either full reel (5 meter) or short reel (1 meter)

Female DC cables (5.5 x 2.1 mm) are pre-mounted on both ends of the reel for quick and easy set up using the PN 3091 power supply.

MECHANICAL DIMENSIONS
(1:4 SCALE)

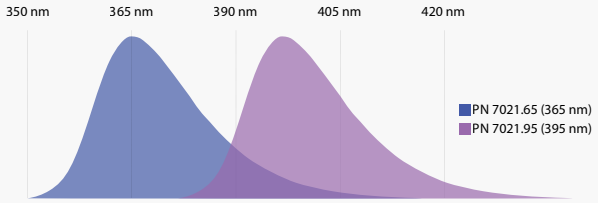
Short Reel: 3.21 ft (1002 mm)
Long Reel: 16.43 ft (5008 mm)

PHOTOMETRIC SPECIFICATIONS

UV output per ft (365 nm):	0.75 W
UV output per ft (395 nm):	1.10 W
Radiometric Efficiency (365 nm):	15%
Radiometric Efficiency (395 nm):	22%
Spectrum FWHM:	10 nm
Emission angle:	120 deg

Download full photometric reports at https://www.waveformlighting.com/photometrics

TYPICAL EMISSION SPECTRUM



We maintain a ±5 nm tolerance for wavelength specifications. FWHM stands for Full-Width Half-Max, the size of the wavelength range across which irradiance is measured to be at least 50% of the peak wavelength irradiance value.

ELECTRICAL SPECIFICATIONS

Input type:	Constant Voltage
Input voltage:	12V DC
Current draw per ft:	400 mA @ 12V DC
Current draw per full reel:	6.0 A @ 12V DC
Power draw per ft:	4.8 W @ 12V DC
Power draw per full reel:	72 W @ 12V DC
Max run:	16.4 ft (5 meters)

MECHANICAL SPECIFICATIONS (FULL REEL)

Length:	16.43 ft (5008 mm)
Width:	0.394 in (10 mm)
Height:	0.067 in (1.7 mm)
LED spacing (OC):	0.327 in (8.3 mm)
Cut-line spacing:	0.984 in (25 mm)
PCB copper thickness:	4 oz
Connection (both ends):	Female DC 2.1 x 5.5 mm connector

POWER SUPPLY SELECTION

Compatible with Waveform Lighting PN 3091 or third-party 12V DC constant voltage power supply. If you choose to utilize a third-party power supply unit, you will need to ensure that the power capacity of the power supply is sufficient for the length of LED strip being connected. Use the table below to determine if the power supply is sufficient for your project.

LENGTH	MINIMUM POWER SUPPLY CAPACITY
1 ft:	420 mA / 5 W
3 ft:	1.3 A / 16 W
6 ft:	2.5 A / 30 W
9 ft:	3.8 A / 46 W
12 ft	5.0 A / 60 W
16.4 ft	6.7 A / 80 W

LENGTH	MINIMUM POWER SUPPLY CAPACITY
0.5 m:	720 mA / 9 W
1.0 m:	1.4 A / 17 W
2.0 m:	2.9 A / 35 W
3.0 m:	4.3 A / 52 W
4.0 m:	5.8 A / 69 W
5.0 m:	7.2 A / 86 W

COMPATIBLE ACCESSORIES

Power Supplies:	3091, 3093.120†
Connectors:	3070, 3071, 3072, 7098, 7094, 7095
Dimmers:	3081, 3093.120 + TRIAC wall-dimmers§
Aluminum Channels:	3060, 3061; diffusers are not compatible with UV output

† Requires PN 7095 or equivalent adapter to connect
§ See tested dimmer list under PN 3094 for additional details

THERMAL MANAGEMENT

Max Ambient Temp (T_A):	125°F (50°C)
Max Case Temp (T_C*):	185°F (85°C)
Typical temp rise:	Δ54°F (Δ30°C)

These LED strip lights are designed to be operated without the need for any additional thermal management. Aluminum channel accessories may assist somewhat in dissipating heat away from the LED strip lights, but are not necessary.

*T_C refers to the temperature of the solder joint between the LED and circuitboard. For non-typical installations where power or thermal density may be higher, monitor this T_C temperature point and verify that the LED solder joints remain below 185°F (85°C) after the system reaches thermal stability.

PART NUMBERS AND ORDERING

365 nm short reel:	7021.65
395 nm short reel:	7021.95
365 nm full reel:	7021.65.5M
395 nm full reel:	7021.95.5M

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waveform lighting

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